No. of Functional Features Included in the Solution

Date	7 July 2026
Team ID	XXXXX
Project Name	Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Functional Features Overview:

Functional Features

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	Home Page	Displays project overview, features, and navigation to the prediction page.	Frontend (Home Page)	Done	High
2	Credit Card Eligibility Form	Allows users to enter applicant details required for prediction.	Frontend (Prediction Form)	Done	High
3	Input Validation	Validates user inputs on both client-side and server-side before prediction.	Frontend & Flask Backend	Done	High
4	Data Preprocessing	Cleans, encodes, scales, and prepares user input before passing it to the machine learning model.	Preprocessing Module	Done	High

5	Machine Learning Prediction	Uses the trained XGBoost model to predict whether the applicant is eligible or not eligible.	Prediction Engine	Done	High
6	Result Generation	Displays eligibility status, confidence score, risk level, and recommendations.	Result Page	Done	High
7	Model Evaluation & Comparison	Compares Logistic Regression, Decision Tree, Random Forest, and XGBoost to select the bestperforming model.	Model Training Module	Done	Medium
8	Cloud Deployment	Deploys the Flask application on Render for online access and testing.	Deployment Module	Done	Medium

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Feature Summary

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Home Page, Prediction Form, Result Page

2	Backend / Logic	3	Input Validation, Data Preprocessing, Prediction Engine
3	Data / Model Storage	1	CSV Dataset, Trained Model (.pkl), Scaler, Encoder
4	Machine Learning	1	Model Training, Evaluation, Model Comparison, XGBoost Prediction
5	Security / Validation	1	Client-side & Server-side Input Validation